

Rohan Kulshrestha

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EDUCATION

Nanyang Technological University

Bachelor of Engineering in Computer Engineering (Expected May 2028)

Singapore

Aug 2023 – Present

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, C

AI/ML: LLM Fine-Tuning (QLoRA, DPO, GRPO), Quantization, Evals, AI Agents & Tool Calling, MCP, LangGraph

Systems, Web and Audio: Real-time DSP, JUCE, CRDTs (Yjs), WebSockets, Next.js, React, React Native, FastAPI

Data, Cloud and Tools: PostgreSQL, DuckDB, AWS (Lambda, DynamoDB), Google Cloud, Docker, Git, Linux

EXPERIENCE

AI Engineer Intern

Jun 2026 – Aug 2026

HoshoAI

Singapore

- Shipped a prompt-optimization service now used by customers: an AWS Lambda backend that scores and improves AI agent prompts via the Claude and OpenAI APIs.
- Developed eval harnesses measuring output quality and run-to-run consistency, with A/B tests before shipping.
- Shipped an MCP integration, also in customer use, that lets a user's coding agent (Claude Code, Codex) join their live editor session as a collaborator, following session events and proposing edits.

Software and AI Engineer Intern

May 2025 – Aug 2025

AlphaGeo | [Writeup](#)

Singapore

- Built the pipeline behind a data center database AlphaGeo now sells to clients: 11,000+ facilities consolidated from 6+ sources (Wikidata, OpenStreetMap, PeeringDB, news, reports) with geospatial and fuzzy-match deduplication.
- Developed a LangGraph agent (Gemini 2.5 Pro) that finds market reports on its own, plus a vision pipeline that extracts data from web-only PDF viewers.

PROJECTS

QuantLab - Quant Research and Trading Stack | Python, C++, LLM Fine-Tuning | [Writeup](#) Jul – Aug 2026

- Created an app that turns a plain-English trading idea into a validated, backtested strategy; my fine-tuned 9B translator scores 28/30 vs a frontier model's 29/30.
- Built an analyst agent that answers finance questions via tools over point-in-time SEC EDGAR data (12.5M facts), then distilled it into a 9B open model: eval score 0.79 to 0.90 on 328 held-out questions vs its frontier teacher's 0.91.
- Fine-tuned Qwen3 14B into a memo writer whose every cited number is machine-verified: 99.6% accuracy vs Claude Sonnet 5's 99.8%, then quantized 16GB to 9GB with no measured loss.
- Wrote a deterministic C++20 matching engine (10.4M ops/sec, hash-verified replay) and a risk gateway driving live paper trading at a real brokerage.
- Designed a backtester where lookahead bias is structurally impossible (933 tests), fed by survivorship-free minute bars I decoded from raw IEX exchange data (24,409 symbols).

CareerSphere - AI Career Exploration App | Python, OpenAI, DuckDB | [Writeup](#)

Jun 2026

- Won Champion (1st place)** at the OpenAI x AI Singapore x PyCon SG 2026 Hackathon, in a team of two.
- Turns a resume into a navigable 3D map of Singapore's job market, built on 2,030 official SkillsFuture roles and 88,000+ live job postings.
- Designed it so the data decides and the AI interprets: an OpenAI tool-calling agent orchestrates while deterministic Python owns skill matching, role-fit scoring and gap ranking; deployed on Cloud Run.

Room - Coordination Layer for Coding Agents | TypeScript, MCP, Yjs CRDT | [GitHub](#)

Sep 2026

- Won 3rd place** at the OpenAI x Airwallex x ClickHouse x AI Tinkerers "Agents Leaving the Chatbox" hackathon.
- Built in a team of three: a Claude Code/Codex plugin that lets teammates' coding agents coordinate in real time, sharing uncommitted edits and claiming code so merge conflicts are prevented instead of resolved.
- Agents message, interrupt and wake each other, get warned when a teammate plans to change code they depend on, and test a merge of everyone's uncommitted work before anyone pushes (14 MCP tools over a Yjs CRDT).

jit.ai - Autonomous ML Engineer Agent | Python, PyTorch, RecSys | [GitHub](#)

Aug 2026

- Developed an agent that improves a video-recommendation model unattended, beating the dataset's baseline; code, not the LLM, keeps a change only if its gain clears measured seed noise.

BQST - Audio Mastering Plugin | C++, Real-Time Systems, Concurrency, JUCE | [Writeup](#)

Jun 2026

- Built and shipped a native C++ audio plugin (VST3/AU/Standalone) with a signed, notarized macOS installer.
- Wrote the DSP core on a lock-free, allocation-free real-time path with 8x oversampling; passes strict pluginval.

ADDITIONAL EXPERIENCE & ACTIVITIES

Singapore Armed Forces, Specialist (Jan 2021 – Jan 2023): Led and trained a 40-member artillery unit.

Music Production: Produced, mixed and mastered original songs (2M+ streams); hand-built analog audio gear.